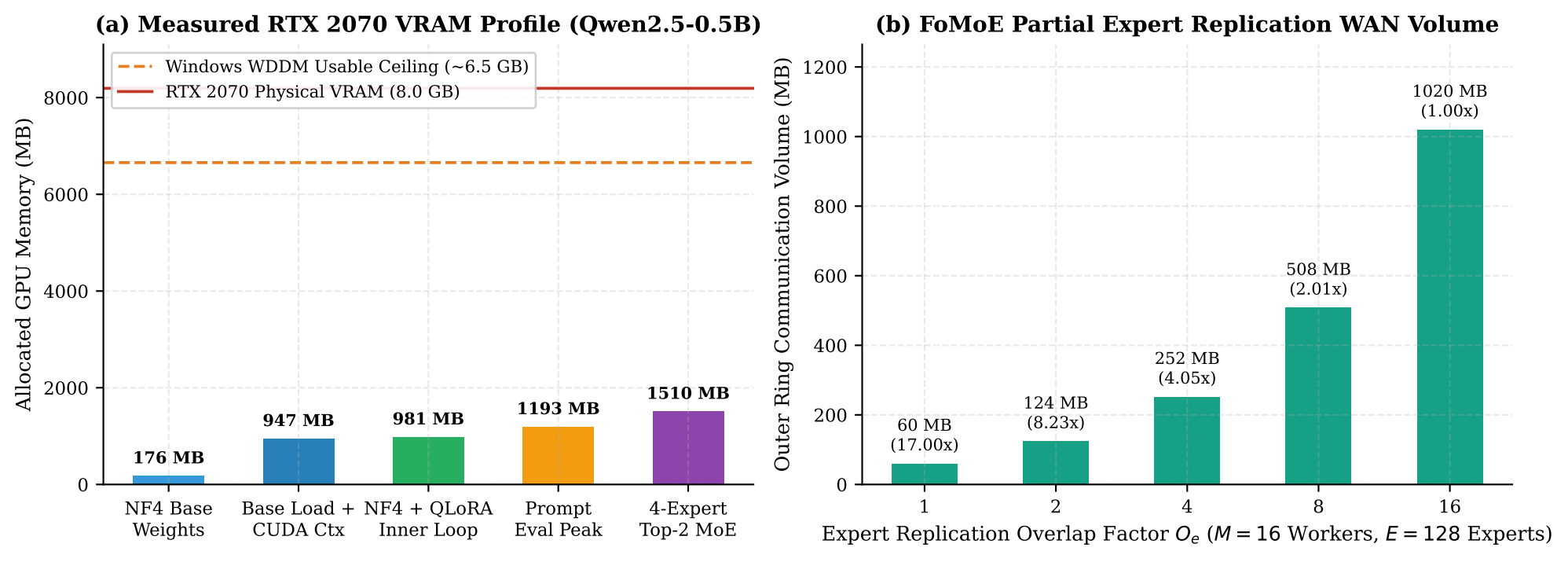
(a) Measured RTX 2070 VRAM Profile (Qwen2.5-0.5B)
Windows WDDM Usable Ceiling (~6.5 GB)
RTX 2070 Physical VRAM (8.0 GB)
Allocated GPU Memory (MB)
0
2000
4000
6000
8000
176 MB
947 MB
981 MB
1193 MB
1510 MB
NF4 Base Weights
Base Load + CUDA Ctx
NF4 + QLoRA Inner Loop
Prompt Eval Peak
4-Expert Top-2 MoE
(b) FoMoE Partial Expert Replication WAN Volume
Outer Ring Communication Volume (MB)
0
200
400
600
800
1000
1200
60 MB (17.00x)
124 MB (8.23x)
252 MB (4.05x)
508 MB (2.01x)
1020 MB (1.00x)
1
2
4
8
16
Expert Replication Overlap Factor O_e ($M = 16$ Workers, $E = 128$ Experts)